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Standard Parameters of the system:

- Time increment  $dt = 10^{-5}s$
- Mass  $M = 0.004kg$
- Radius  $R = 0.005m$
- Grav. acc.  $g = 9.80665m/s^2$
- Outer Magnetic Field  $B_x = B_y = 0, B_z = 5 \cdot 10^{-5}T$
- Initial values  $x_0 = y_0 = 0$
- Rolling Resistance coef.  $\nu = 0.002$
- Mass density of air  $\rho = 1.2kgm^3$
- Drag coefficient of sphere  $c = 0.47$

# 1 Sketches of the System: In Paper

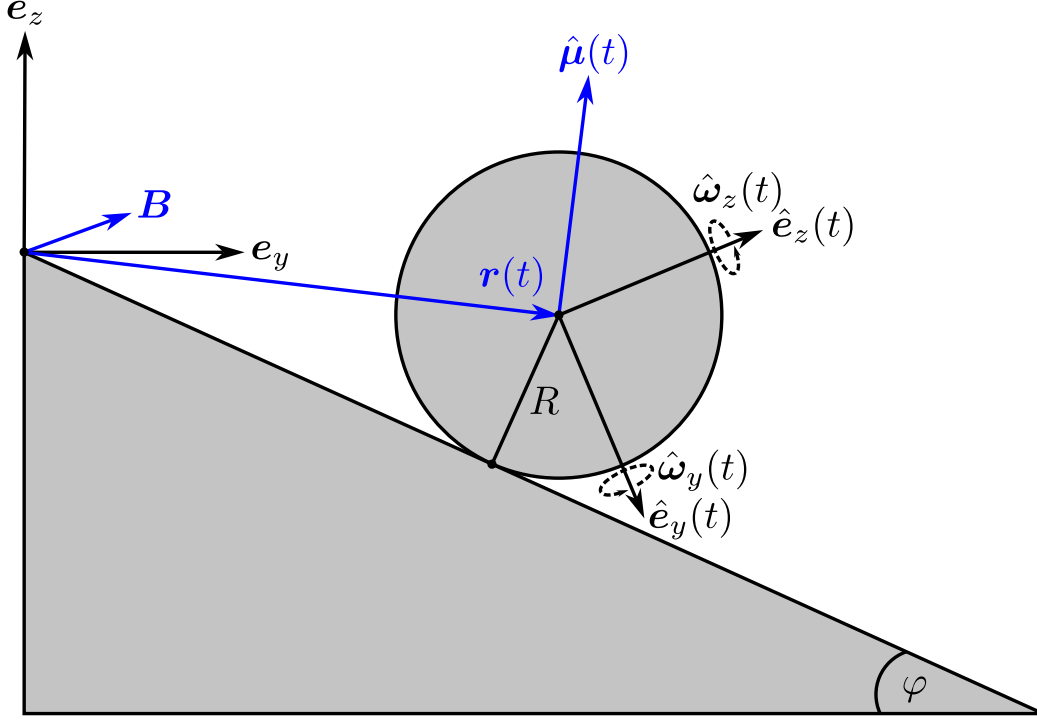


Figure 1: Depicted is the  $yz$ -projection of the incline system, the  $z$ -direction is orthogonal to the paper plane. The incline has an inclination angle  $\varphi$  and the sphere is of radius  $R$ . Two orthonormal coordinate frames are considered: The fixed frame  $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$  and the body frame  $\{\hat{\mathbf{e}}_x, \hat{\mathbf{e}}_y, \hat{\mathbf{e}}_z\}$ . Vectors expressed in the body frame carry hats and are time-dependent, stationary vectors not. Blue vectors might have a non-zero  $z$ -component and are therefore only projections. The outer magnetic field is represented by  $\mathbf{B}$ , the magnetic dipole vector of the sphere by  $\hat{\boldsymbol{\mu}}(t)$ , and the centre of mass measured from the fixed frame by  $\mathbf{r}(t)$ . The in-plane angular velocities of the body frame are shown as  $\hat{\omega}_x$  and  $\hat{\omega}_y$ .

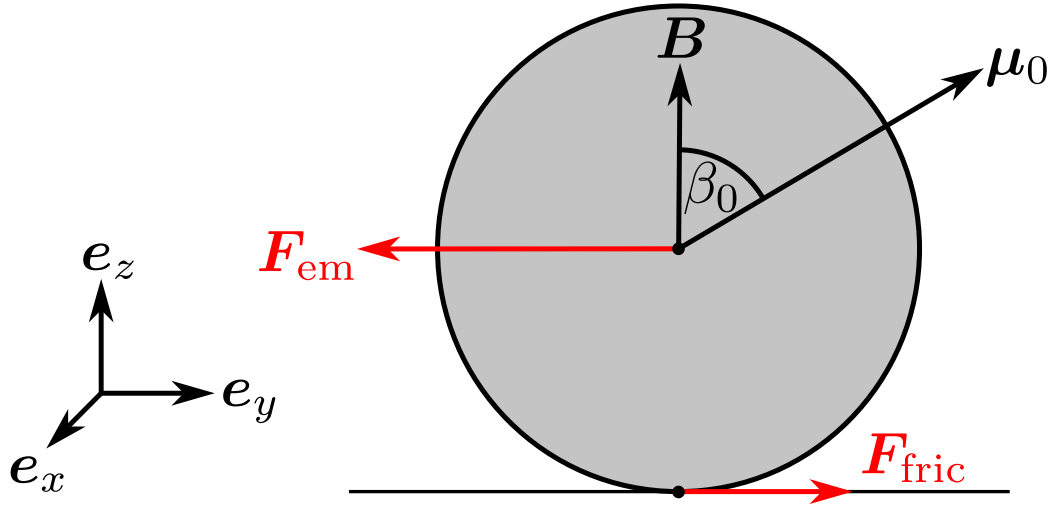


Figure 2: Shown is a sketch of the flat system with sufficiently small initial displacement angle  $\beta_0$  between the outer magnetic field  $\mathbf{B}$  and the initial magnetic dipole vector  $\boldsymbol{\mu}_0$ . Note that the system is effectively reduced to two dimensions. The red vectors indicate the oscillation-driving electromagnetic force  $\mathbf{F}_{\text{em}}$  and the damping frictional force  $\mathbf{F}_{\text{fric}}$ . The frictional force includes aerodynamical drag and rolling resistance.

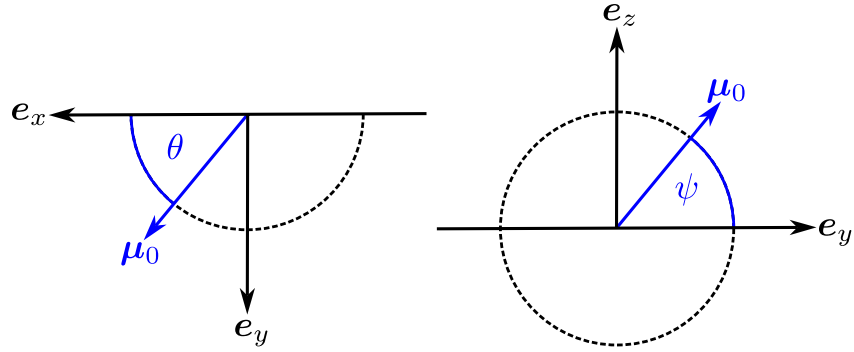


Figure 3: Parametrizations of  $\boldsymbol{\mu}_0$  in F and G.

## 2 Figure Family A: Appendix

These figures are results of a proof-of-concept toy model: The incline without electromagnetic interaction. The goal is to compare numerical results to analytical equations.

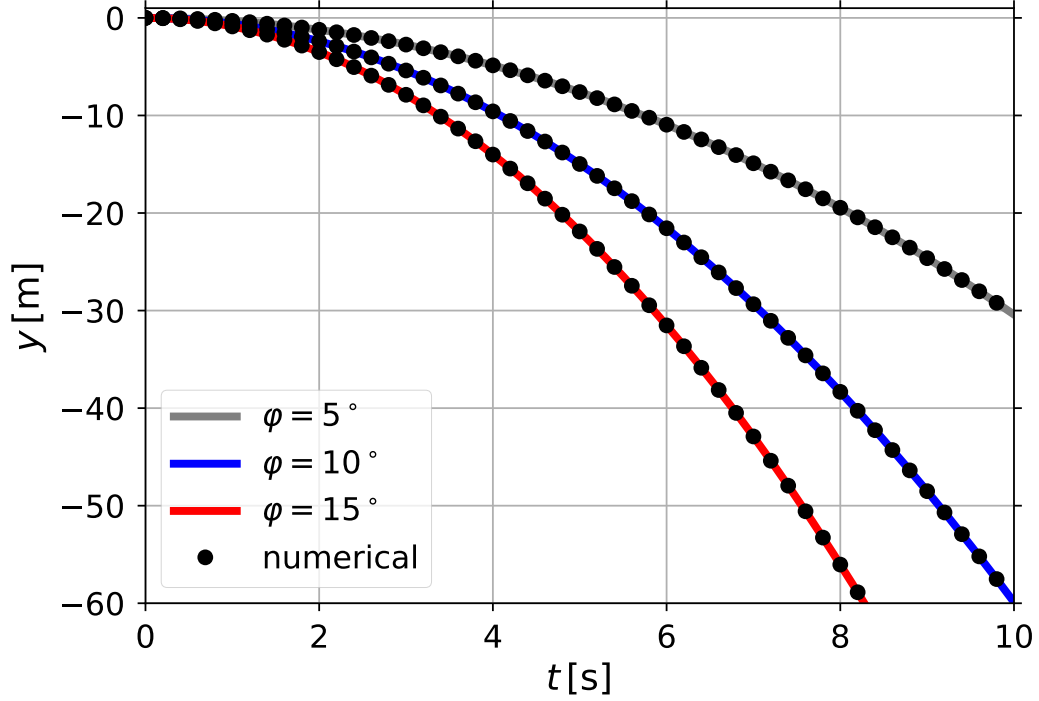


Figure A1: Shown are different evolutions of the  $y$ -projection of the centre-of-mass coordinate  $\mathbf{r} = (x, y, z)^\top$  over time  $t$  for different incline angles  $\varphi$  without friction. Lines were calculated from the analytical solution and compared to the numerical results, represented as dots.

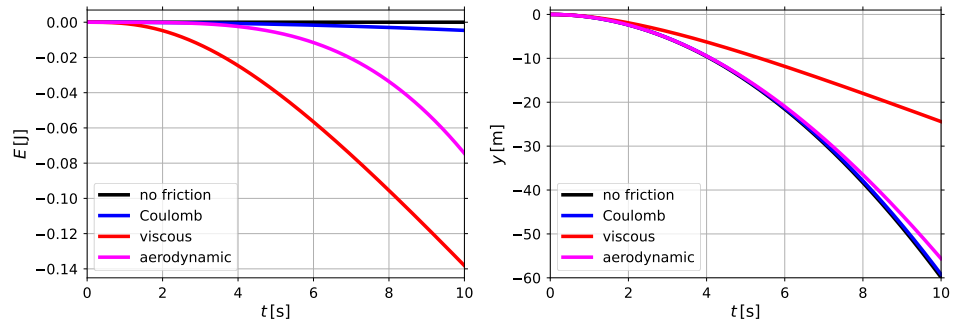


Figure A2/A3: The left plot shows the energy evolution  $E(t)$  of the system over time  $t$ , while the right one depicts the  $y$ -COM-coordinate evolution. The data originates from the numerical simulation program. Compared are different friction models, where *Coulomb* represents dry rolling resistance, *viscous* represents a linear damping approximation to rolling resistance, and *aerodynamic* represents spherical aerodynamic drag.

### 3 Figure Family B

## 4 Figure Family C



## 5 Figure Family F: In Paper

Now combined with former figure J. The fixed non-standard parameters were:

- $\Delta t = 1s$
- $\phi = 10^\circ$
- Rolling Resistance on, Aerodynamical drag on

while the initial magnetic dipole moment was parametrized as

$$\boldsymbol{\mu}(t = 0) = \mu(\sin \varphi, \cos \varphi, 0)^\top$$

with polar angle  $\varphi$  and modulus  $\mu = \|\boldsymbol{\mu}\|$ . For each modulus  $\mu \in \{0.0, 0.1, \dots, 2.5\}$ , the polar angle was varied for  $\varphi \in [-\frac{\pi}{2}, \frac{\pi}{2}]$  in  $\frac{\pi}{399}$ -steps, yielding  $26 \cdot 400 = 10400$  data sets. The goal was to measure the influence of the magnetic moment on the deviation of the trajectory from a straight path.

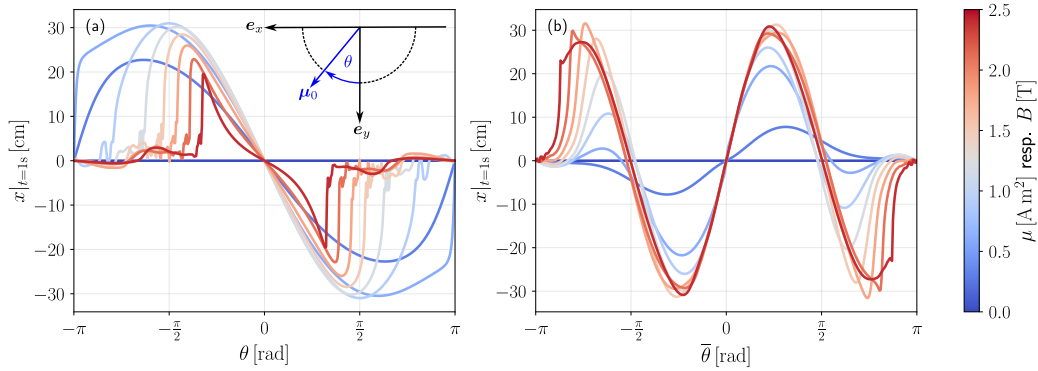


Figure F1: Final  $x$ -deviation w.r.t. polar angle and modulus. Figure (a) and (b) both show the  $x$ -value of a trajectory after  $t = 1s$  depending on the initial polar angle. Each graph represents a modulus of  $\boldsymbol{\mu}$ , indicated by the color bar. The first ten graphs in (a) show that the trajectorial deviation increases non-linearly with the angular misalignment between  $\boldsymbol{\mu}$  and the  $yz$ -plane. As expected, the deviation is antisymmetric about the origin. For increasing values of  $\mu$  and  $\varphi$ , the deviation saturates. In (b), the curves for the remaining moduli  $\mu$  can be seen. In this range, the magnetic forces start to dominate over the gravitational force for high polar deviations. The extrema of the curves correspond to the point where trajectories begin to loop around the origin instead of following the incline.

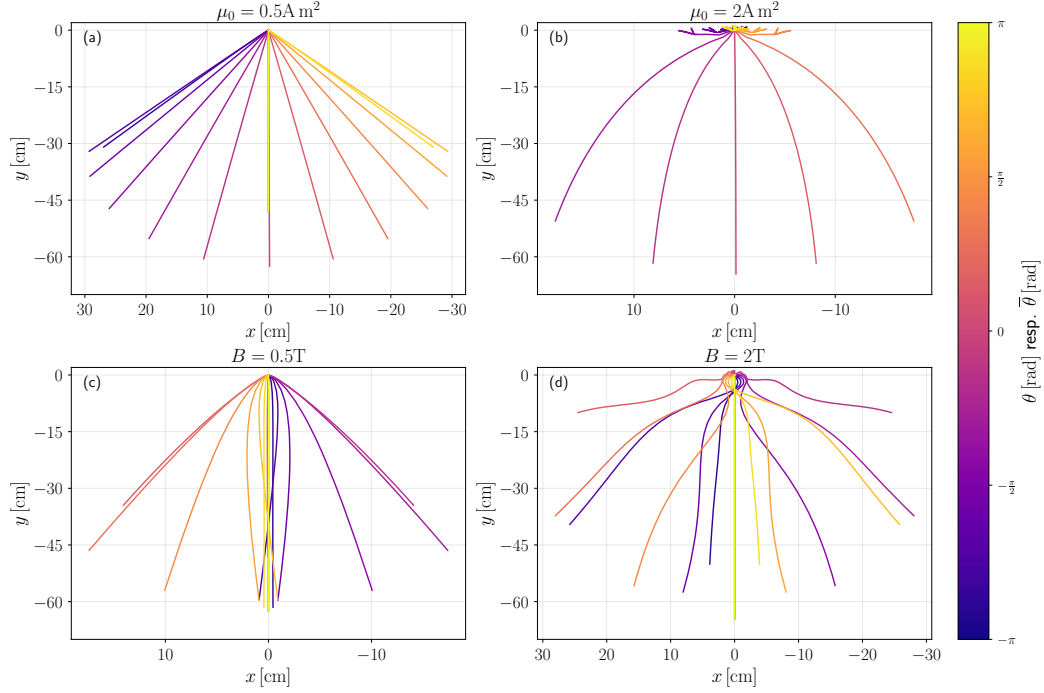


Figure F2: Chosen trajectories for fixed moduli. Figures (a), (b), and (c) show trajectories projected on the  $xy$ -plane for fixed values of  $\mu$  and eleven equally spaced polar angle samples. It can be seen that in all cases, higher angular deviations correspond to stronger deflection of the sphere from a straight path. Increasing the modulus increases the curvature of the trajectories which evolve from almost linear paths to near-circular arcs. Figure (c) also shows the behaviour past the critical points of Figure F1(b) where the outmost paths form closed loops.

## 6 Figure Family G: In Paper

The fixed non-standard parameters were:

- $\Delta t = 1s$
- $\phi = 10^\circ$
- Rolling Resistance on, Aerodynamical drag on

while the initial magnetic dipole moment was parametrized as

$$\boldsymbol{\mu}(t=0) = (0, \sin \vartheta, \cos \vartheta)^\top$$

with polar angle  $\vartheta$  and modulus  $\|\boldsymbol{\mu}(t=0)\| = 1$ . The goal was to analyze at which point the magnetic force, which aligns the magnetic moment with the outer field, is surpassed by gravity.

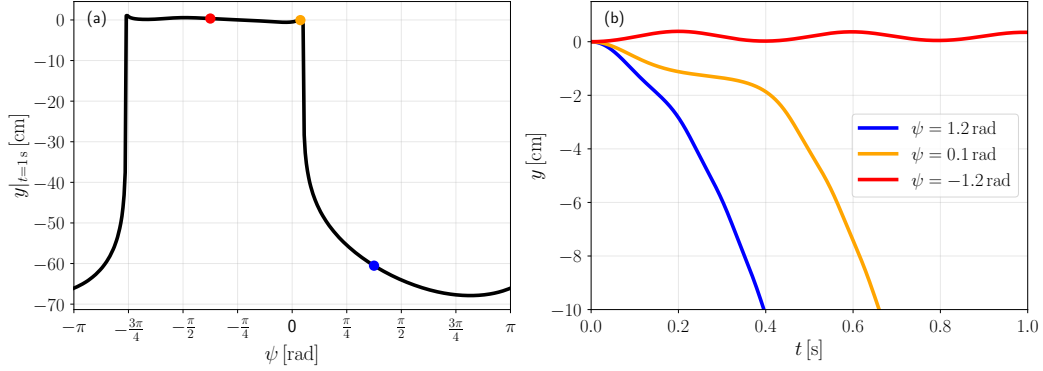


Figure G: Total  $y$ -displacement after  $t = 1$  s for different polar angles (a) and associated representative  $y$ -projected trajectories (b). Figure (a) shows the total distance in  $y$  the sphere has rolled after one second depending on the polar misalignment of  $\boldsymbol{\mu}$  and  $\boldsymbol{B}$ . There are three discernible regions: For almost all positive and a small range of negative deviations, the sphere covers distances characteristic for downward rolling motion. In the negative regime and around zero, the sphere is remaining around the origin. Between these regions, there are two steep boundary regions indicating an abrupt change between downward motion and oscillation around the origin. Figure (b) shows the evolution of the  $y$ -coordinate in time for three representative trajectories as indicated by the points in (a). The non-downward motion corresponds clearly to oscillation around the origin, while the downward motion shows only very weak signs of oscillatory motion in the beginning. The intermediate curve displays strong oscillatory motion in the beginning which is then overcome by gravity.

## 7 Figure Family H

These figures were generated with the following non-standard parameters:

- $\Delta t = 10s$
- $\phi = 5^\circ$
- Rolling Resistance on, Aerodynamical Drag on

The magnetic moment at  $t = 0$  was again parametrized by

$$\boldsymbol{\mu} = \mu \cdot (\sin \varphi, \cos \varphi, 0)^\top.$$

The goal was to observe the behaviour of the magnetic moment vector in the rotating frame of the sphere. We considered two cases: Downward rolling motion for  $\varphi = \frac{\pi}{5}$  and  $\mu = 0.1Am^2$  and looping motion for  $\varphi = \frac{\pi}{2}$  and  $\mu = 1.5Am^2$ . We lowered the slope and  $\mu$  for the first one to obtain slower rotational velocities, making trajectories slightly easier to interpret. For the plots of  $\boldsymbol{\mu}$  in time, we decomposed

$$\boldsymbol{\mu} = \boldsymbol{\mu}_{\parallel} + \mu_{\perp} \mathbf{e}_z.$$

The drawn out trajectories correspond to the  $\mu_{\parallel}$ -component, i.e. the projection of  $\boldsymbol{\mu}$  onto the  $xy$ -plane. The origin of the projected  $\boldsymbol{\mu}(t)$  is the centre of mass of the sphere at time  $t$ . The trajectories of the COM were also plotted in arbitrary units, since they have to be scaled up to be visible in relation to the length of  $\boldsymbol{\mu}$ .

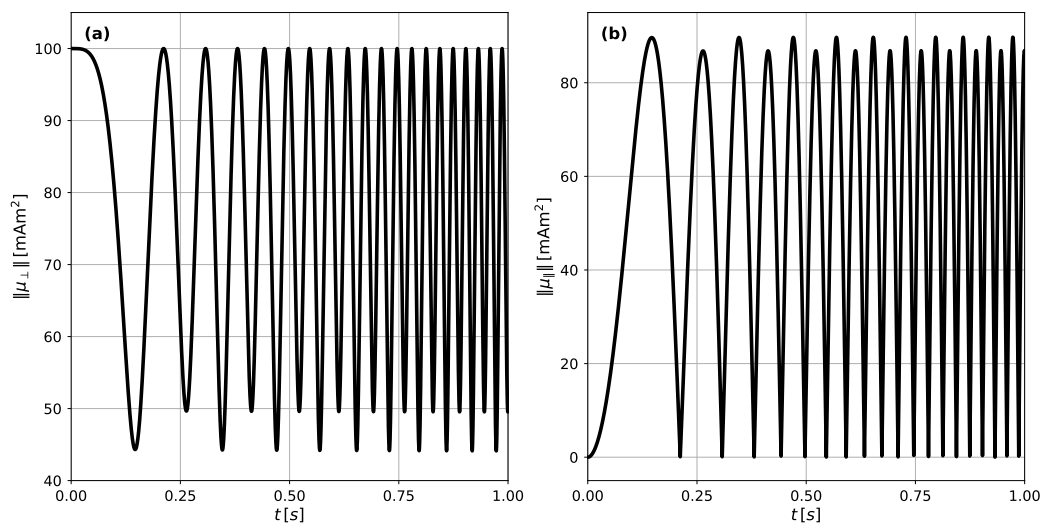


Figure H1:

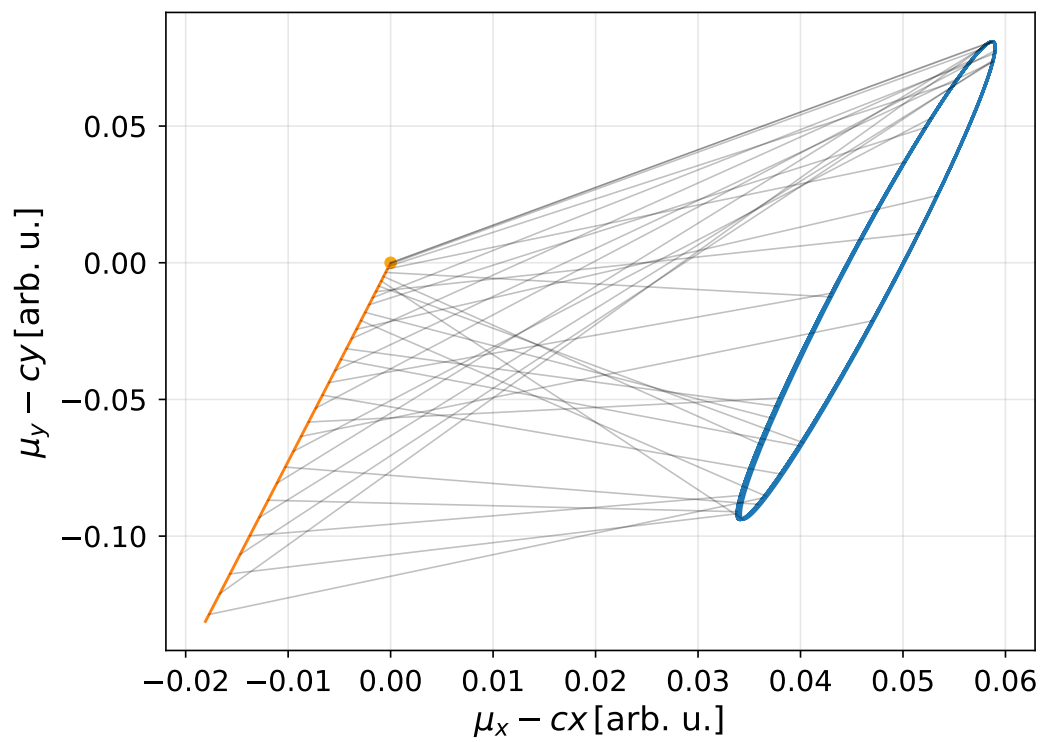


Figure H2:

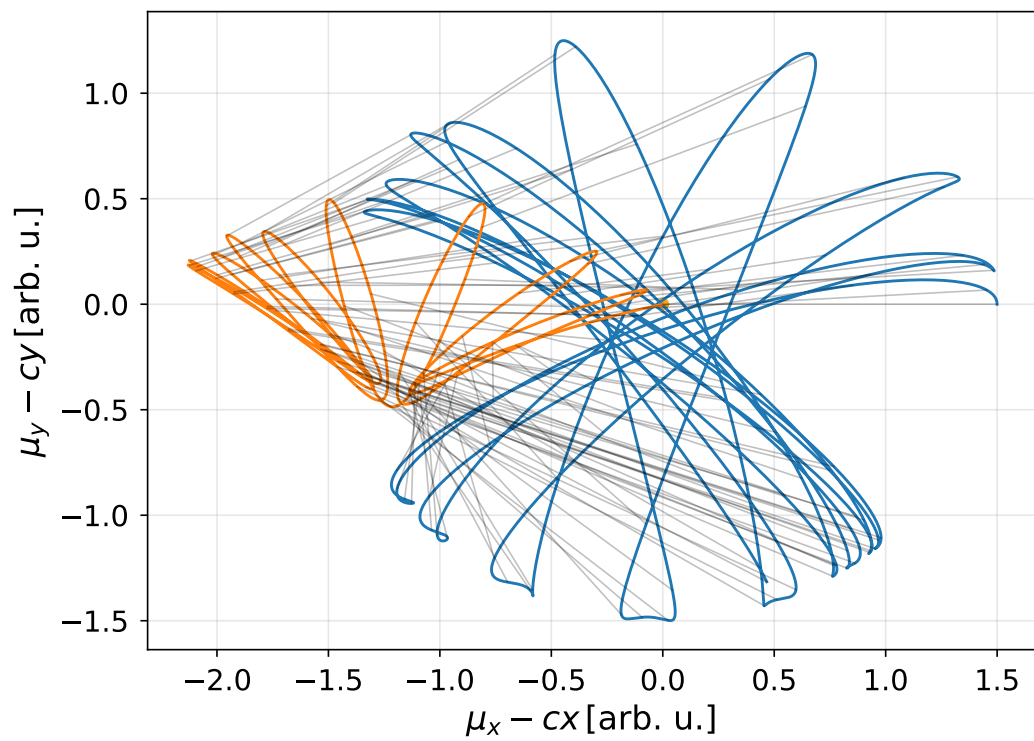


Figure H3:

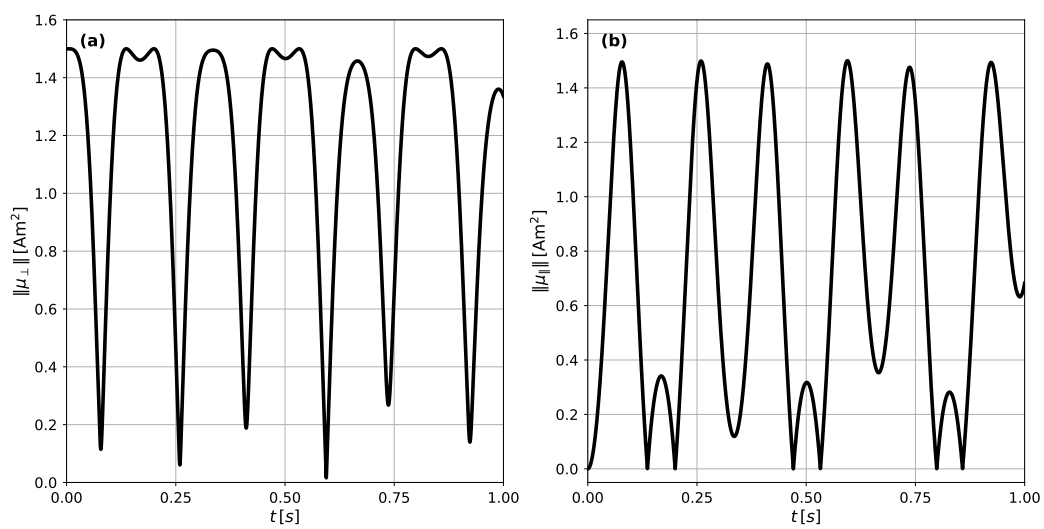


Figure H4:

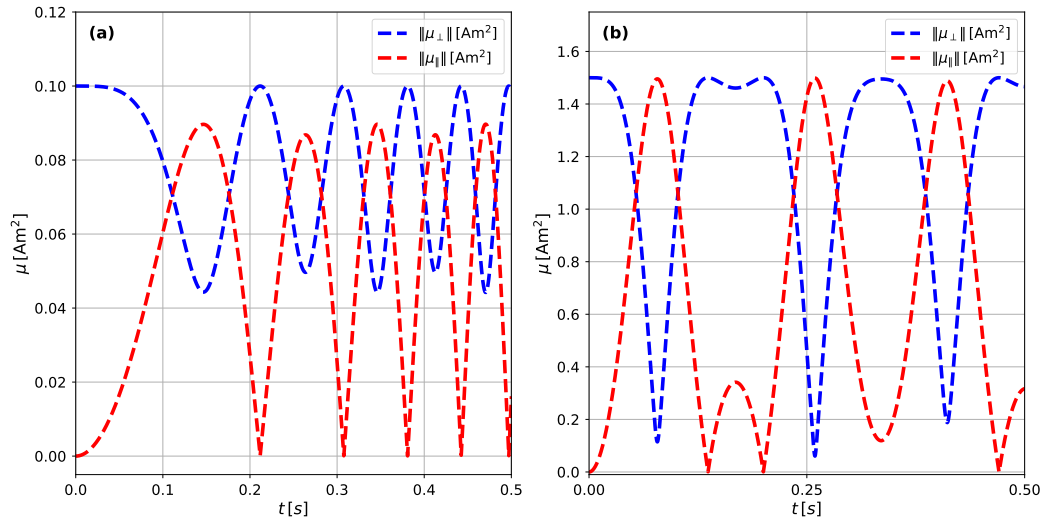


Figure H5:



## 8 Figure Family I: Not for paper

The non-standard parameters are

- $\Delta t = 10s$
- $\phi = 30^\circ$
- Rolling Resistance on, Aerodynamical Drag on

Goal of these calculations was to take a closer look at which inputs are close in orders of magnitude to the observed motion of the sphere. The idea is to assume that  $\mathbf{B} = B_z \mathbf{e}_z$  and  $\boldsymbol{\mu}(t=0)$  are almost perfectly aligned at rest, with a little polar perturbation in the  $xy$ -plane causing deflection of the path. To quantify the size of the perturbation, we considered  $\varphi \in \{0, \frac{\pi}{20000}, \frac{\pi}{2000}, \frac{\pi}{200}, \frac{\pi}{20}, \frac{\pi}{2}\}$ .

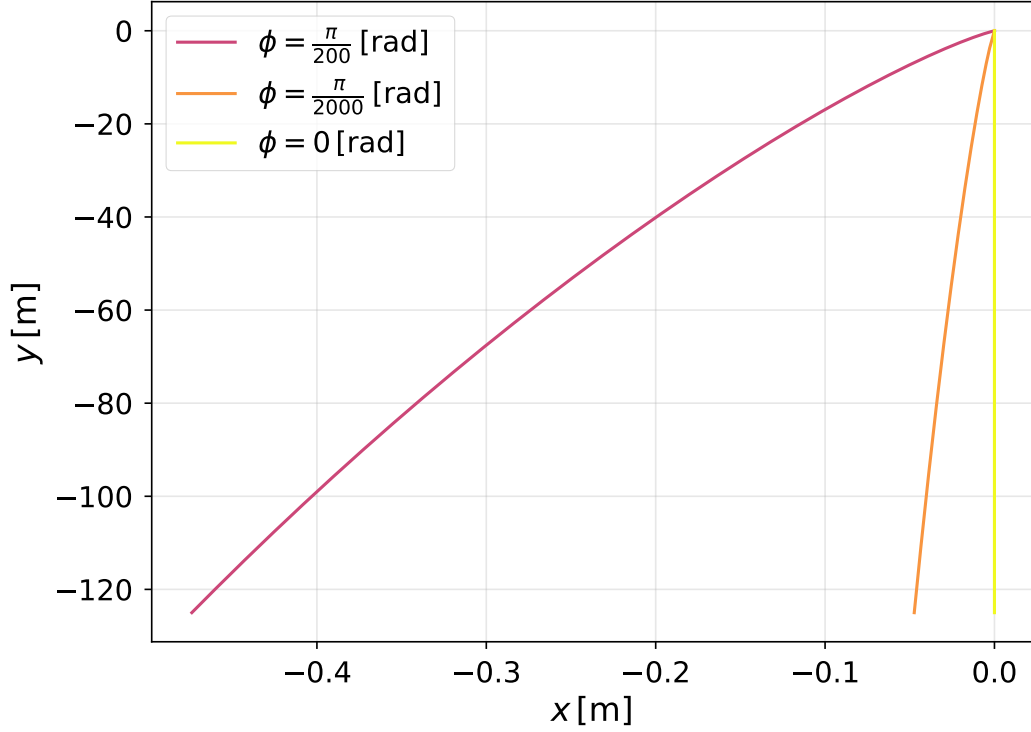


Figure I1: The paths show that only  $\frac{\pi}{200}$  is in the correct range to be close to the experiment quantitatively.

## 9 Figure K: In paper

The non-standard parameters are

- $\Delta t = 1s$
- $\phi = 20^\circ$
- RR on, AD on
- $\mu_0 = 1.0Am^2$
- No initial deviation: If the moment and field are not coplanar with the rolling direction, the rolling causes the misalignment after a few time steps automatically.

The goal is to investigate the general behaviour of the ball in roughly realistic settings. We used the NCEI model to derive the magnetic field vector at  $53.5501721^\circ$  north and  $10.0013165^\circ$  east (Hamburg, Centre) with a mean sea level elevation of  $25m$  at the 12th of May, 2026. The total magnitude was rounded to  $5 \cdot 10^{-5}T$ , close to the actual  $50159.3nT$  of the NCEI. The magnetic field in the East-North-Up-Frame is then given by

$$\mathbf{B} = B \cdot (\cos I \sin D, \cos I \cos D, -\sin I),$$

where the NCEI predicts an inclination of  $I \approx 68.6^\circ$  and a declination of  $D \approx 4.2^\circ$ .<sup>1</sup>

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<sup>1</sup>The geographical spherical system is quite counterintuitive from the phycisists standpoint: The inclination is the dip of the field, so  $I > 0$  means that the field points into the ground.

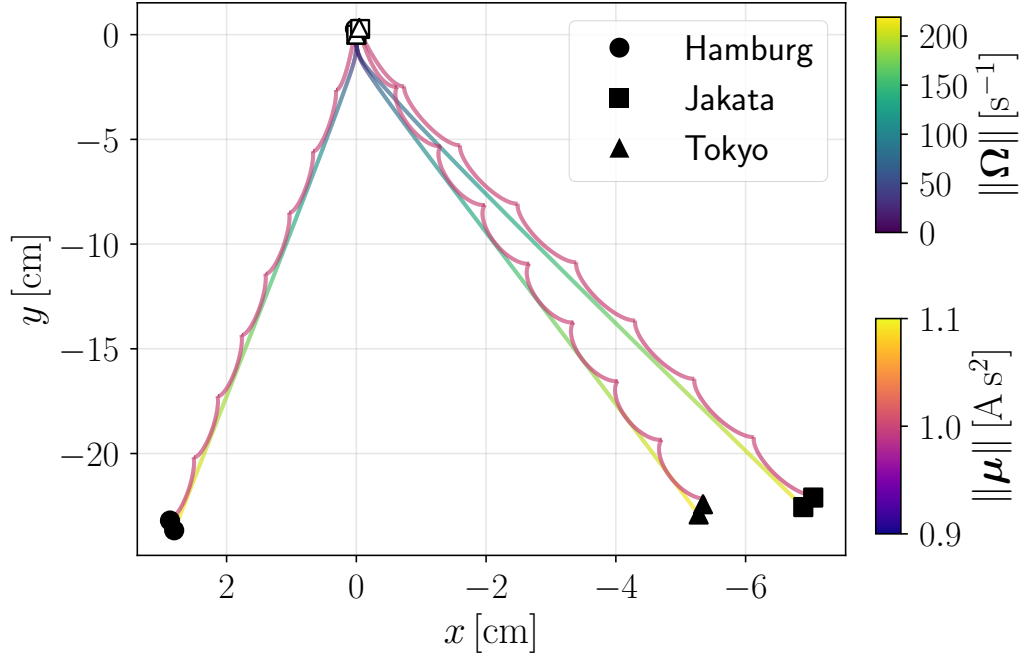


Figure K1: Shown is the trajectory in the  $xy$ -plane and the evolution of the magnetic moment vector's direction (i.e. a point on the sphere in the direction of  $\boldsymbol{\mu}$ ).